Project Proposal: Bayesian Estimation

Stella Rovazzi · Ramez Elprince · Brenden Doogan

University of Guelph

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1 Project Overview

maybe quick paragraph here

Objectives

- bleh [4]

2 Background and Literature

Bayesian Data Analysis, Gelman et al. [2]

3 Proposed Computational Strategy

3.1 Quantities of Interest

- Estimated parameter mean and variance/standard deviation
- credible intervals (informs us of the posterior model) [2]
- Gelman-Rubin Statistic (Convergence Assessment) [5]
- Rate of Convergence (/Assessing with Gibbs Stopper) [6]
- Warm-up range
- Maximum a priori

3.2 Required Libraries

- MASS for the bivariate normal generation (or basic R) [4]
- STAN for the actual MCMC/Gibs sampling – we may some examples by hand [3]

3.3 Simple Example

Goal

Ensure we can converge to a known model using bayesian techniques.

Methodology

This follows the example C.3 regarding the normals in Bayesain Data Analysis (BDA) [2].

1. Generate bivariate normal data
2. Derive form of marginals and implment as R functions (assuming normal prior and conditional)
3. Sample from posterior with Gibbs Sampler
4. Assess the if sampled posterior is reasonable
5. Retrieve the other quantites of interest

3.4 Harder Example

Goal

Apply what was learned in the simple example to real data. We will be able to try and compare the our methodology.

Methodology

This follows the bioessay expirement in example 3.7 from BDA [2].

1. Derive the full conditional, then the marginal conditionals using the distributions provided in text
2. Sample from posterior with Gibbs Sampler
3. Assess the if sampled posterior is reasonable
4. Retrieve the other quantites of interest

Dataset

This dataset is shown in table 3.1 from the BDA text [2]. It has been replicated below.

Bioassay data from Racine et al. (1986), copied from BDA

Dose, x_i (log g/ml)	Number of animals, n_i	Number of deaths, y_i
−0.86	5	0
−0.30	5	1
−0.05	5	3
0.73	5	5

The data is binomially distributed.

3.5 Extend to Hierarchical Models

Goal

If time permits, we would use the rat tumor example (5.4) from BDA and tutorial C.3 to try to construct a bayesian hierarchical model using stan [2], giving us opportunity to learn hyperparameters and Stan.

Methodology

1. Create stan file, laying out parameters, model, and transformed parameters
2. Run some number of chains iteratively in R, which can be set to be the Gibbs method
3. Use Stan and R to find the posterior
4. Assess the if sampled posterior is reasonable
5. Retrieve the other quantites of interest

Dataset

This dataset can be found at Gelman [1] titled "Rat tumors". The data represents the count of rats developing tumours from various expirements (or samples). Y representing the number of rats developing tumors and N representing the total number of rats in the study. The data is binomially distributed.

Division of Responsibility

Timeline

We can chat about GANTT charts on tuesday or smth, just so we can layout realistic dates with buffer room.

Expected Outcomes

bleh

References

[1] Andrew Gelman. Data from examples in bayesian data analysis, n.d. URL `https://sites.stat.columbia.edu/gelman/book/data/`.

[2] Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian data analysis*. Texts in statistical science. CRC Press, Taylor & Francis Group, Boca Raton, third edition edition, 2014. ISBN 978-1-4398-4095-5 978-1-4398-9820-8.

[3] Jiqiang Guo, Jonah Gabry, Ben Goodrich, Andrew Johnson, Sebastian Weber, and Hamada S. Badr. rstan: R Interface to Stan, July 2015. URL `https://CRAN.R-project.org/package=rstan`. Institution: Comprehensive R Archive Network Pages: 2.32.7.

[4] Brian Ripley and Bill Venables. MASS: Support Functions and Datasets for Venables and Ripley's MASS, May 2009. URL `https://CRAN.R-project.org/package=MASS`. Institution: Comprehensive R Archive Network Pages: 7.3-66.

[5] Maria L. Rizzo. *Statistical computing with R*. CRC Press, Taylor & Francis Group, Boca Raton, second edition edition, 2019. ISBN 978-1-4665-5332-3 978-1-4665-5333-0.

[6] Martin A. Tanner. *Tools for Statistical Inference: Methods for the Exploration of Posterior Distributions and Likelihood Functions*. Springer Series in Statistics. Springer New York, New York, NY, 1996. ISBN 978-1-4612-8471-0 978-1-4612-4024-2. doi: 10.1007/978-1-4612-4024-2. URL `http://link.springer.com/10.1007/978-1-4612-4024-2`.